

Rapid Application Development with

QCUBED



This is intended to be the first in a series of articles on rapid Web application development using the QCubed PHP development framework. The series is intended for new to intermediate-level PHP developers, to guide them on how to develop Web applications faster with QCubed.

It goes without saying that Web application development is a skill that is in great demand, as well as a hot passion among those enchanted by the magic they see on the Web! However, it takes a lot of time and effort. For instance, if you want to build a blog from scratch, let's look at what you require:

1. A database—let's say PostgreSQL or MySQL
2. The LAMP or LAPP stack
3. Tables in the database to hold posts, comments and your login system, so that no one but you can add blog posts.

The first couple of requirements are not hard; all they need are the downloading and installing of packages. The tough part is writing SQL queries, designing data entry forms and so on. You may say it shouldn't take more than six hours—but what if I said that you could do it in just one hour? I know, that might sound a bit far-fetched—but it's reality! And what about adding more functionality to let others make blog posts? What about allowing them to register? And while building websites, we often strive to improve them. If you think the improvements mentioned could take a day: creating new columns, modifying queries and testing them—you could be right. However, you can also get it done in about half the time. Allow me to show you precisely *how*...

Meet QCubed

QCubed is a PHP development framework like CodeIgniter, CakePHP, Symfony, etc. It facilitates rapid Web development. However, there are a few good things about QCubed that give it an edge over other frameworks; things that make it a true RAD (Rapid Application Development) tool. Each feature is unique, and I will cover them one at a time.

Easy to set up

Installing QCubed is simple. You extract files, copy them to a location under your webroot directory, modify the configuration file, and you are ready to go!

Database support

QCubed has support for MySQL, PostgreSQL, Oracle, Microsoft SQL Server, Informix and DB2. This means you can use the framework no matter which database you want to use. Apart from this, you can easily use more than one database in your application.

Code generation

Code generation (or codegen) is the most distinguishing feature of QCubed. It is this one feature that makes your development process so very fast, by removing a large part of the tedious, mundane tasks you have to do when developing. You should be excited—there are multiple things that codegen brings to the table.

One of its major advantages is that it minimises the amount of SQL queries you have to write, and all the routine tasks that come with each (parsing the results for SELECT queries, writing queries to insert or update data, and so on). QCubed achieves it by creating individual classes for each table. You can access the rows of the database as objects of that class. To get the columns of a row, you access the properties of the row object. It also generates functions for selecting data from the table, which create SQL queries automatically; you get results as an array of objects, which is a lot easier to play around with.

In real life or databases, handling relationships is a pain when there are too many! The codegen process will create functions for handling relationships for each of your tables. It will consider your foreign keys and will allow you to query tables using those methods; they too are generated automatically! So loading the blog post of a comment is actually just one more word at the distance of an arrow mark, that's it!

Indexes have a reason to exist—they make queries run faster at the database level. QCubed's code generator acknowledges this, and creates special functions for columns on which indexes are defined. This makes sure that you